Sentinel AI - Client Architecture Brief

AI-powered service health and root-cause analysis for AWS data pipelines

- Audience: client architects and platform leaders
- Scope: live capabilities, architecture fit, and measurable impact
- Version date: 2026-07-08

Problem We Solve

Current incident analysis is manual and slow

- Engineers manually query failures, build log queries, and inspect hundreds of lines

- Root cause write-up and stakeholder summary are done separately

- Recurring incidents are not consistently linked to prior cases

- Result: high mean-time-to-root-cause and high dependence on senior experts

What The Application Delivers

Two-step workflow with guided AI analysis

- Step 1: Fetch FAILED records by time range and component
- Step 2: Analyze selected records using LLM + log backend queries
- Output includes grouped failure patterns, root cause, likely fix, and escalation path
- One-click deep-link opens the exact CloudWatch/Grafana query and window

Core Features

Production-focused capabilities

- Paginated failure table with cross-page selection and bulk analyze
- Executive summary generated for ticket/incident communication
- Recurring issue detection via root-cause signature and CRM case linking
- Pluggable providers for data source, LLM, and log backend
- GitHub MCP code-change context for failures in the selected time window

Architecture Fit

Composable design for enterprise environments

- Frontend SPA + FastAPI backend + provider strategy interfaces
- Backend isolates DataSource, LLM, and LogAnalysis with plugin registry
- Provider swap via configuration, without orchestration code changes
- Postgres persistence stores case history and recurrence signals

Business Advantages

Why this is valuable for platform operations

- Faster incident diagnosis and higher on-call confidence
- Standardized RCA quality across teams and shifts
- Reduced reliance on tribal knowledge and ad-hoc runbooks
- Higher transparency: repeat failures become visible and actionable

Time Effect

Indicative effort comparison per incident

- Manual baseline: 60-180 minutes (querying, log digging, report writing)

- With Sentinel AI: 2-10 minutes (fetch, analyze, review output)

- Estimated reduction: 85%-98% in time-to-root-cause

- Assumption: read-only cloud access and configured providers are available

Adoption Path

Low-friction rollout plan

- Pilot with one data pipeline profile and one operations squad
- Track MTRC, recurrence closure rate, and false-positive trend
- Expand to additional services and teams after pilot baseline
- Roadmap: deeper automation and broader source-code intelligence